

Time-dependent Hubbard magnetism and light-driven spin currents in graphene nanoflakes

Implementation notes, figures, and physics summary

Working report

July 9, 2026

Abstract

This report documents a set of changes to the tight-binding code for graphene nanoflakes, in the order they were built. Three capabilities are covered: (i) the self-consistent Hubbard (UHF) *static* magnetic ground state that underlies everything — the “frozen” field; (ii) its *time-dependent* (TDUHF) extension that lets the magnetic order respond to an optical drive; and (iii) a per-bond spin-current output that makes the spin flow visible in real space. We explain the physics of each, the exact mean-field decomposition used to avoid double counting, the numerical instability that a naive dynamic implementation triggered and how it was cured, the configuration switches, and the four analysis figures produced, with a reading of what each one tells us. Throughout, the reference system is the 5×5 zigzag-edge triangular flake (46 sites, sublattice imbalance $N_A - N_B = 4$, Lieb spin $S = 2$).

Contents

1	Background and starting point	1
2	Implementation 1: static (frozen) Hubbard/UHF magnetism	1
2.1	Motivation	2
2.2	The Hubbard interaction at mean-field level	2
2.3	Self-consistency algorithm (<code>solve_hubbard_mft</code>)	2
2.4	Where the field lives (and a subtlety fixed during development)	2
2.5	The value of U	3
2.6	Configuration, outputs, validation	3
3	Implementation 2: time-dependent (TDUHF) spin channel	3
3.1	Goal and the key physical constraint	3
3.2	Charge/spin decomposition (why there is no double counting)	3
3.3	What is added each step	4
3.4	The instability that a naive version triggered, and the cure	4
3.5	Validation	4
3.6	Configuration and files	4
4	Implementation 3: per-bond spin current	5
5	What we measure, and the two drives	5
6	Figures and what they tell us	5
6.1	Control / validation: <code>plot_hubbard_dynamics.py</code>	5
6.2	The U_{spin} dial: <code>plot_uspin_dial.py</code>	5
6.3	The response matrix: <code>plot_response_matrix.py</code>	6
6.4	Light-driven spin current in real space: <code>plot_spin_current_realspace.py</code>	6

7	Physics findings	6
7.1	The gap and why the resonance moves so much with U	6
7.2	Collective spin mode	7
7.3	Spin transport at fixed S_z	7
8	How to reproduce	8

1 Background and starting point

The code evolves a spinful single-particle density matrix $\rho(t)$ (site basis, ordering $[\uparrow_0 \dots \uparrow_{N-1}, \downarrow_0 \dots \downarrow_{N-1}]$) under a von Neumann equation with a phenomenological relaxation $-\frac{\gamma}{2}(\rho - \rho_0)$. Interactions enter at mean-field level:

- a **charge** Hartree term $V_{ee}(\rho - \rho_0)$ that is spin-blind and vanishes at equilibrium (it acts only on the *induced* density);
- a self-consistent **Hubbard (UHF)** onsite exchange field that produces the magnetic ground state. In the particle-hole-symmetric form the onsite potential for spin σ is

$$V_{i\sigma} = U(\langle n_{i,-\sigma} \rangle - \frac{1}{2}). \quad (1)$$

The history below has three steps: the Hubbard field is first solved once and held **frozen** to give the magnetic ground state (§2); it is then made **time-dependent** so the spin responds to the drive (§3); and finally a **per-bond spin current** is added to see the resulting spin flow in real space (§4).

2 Implementation 1: static (frozen) Hubbard/UHF magnetism

This is the foundation: the self-consistent magnetic ground state that all the later work sits on. It was added first, and is “frozen” in the sense that it is solved once and held fixed during the time evolution.

2.1 Motivation

Before this feature the model had **no equilibrium magnetism**. The up and down blocks of H_{TB} were identical, so every eigenstate was spin-degenerate, $\langle n_{i\uparrow} \rangle = \langle n_{i\downarrow} \rangle$ at every site, and the net moment was exactly zero. The only spin signal in the whole simulation was the tiny ($\sim 10^{-9}$) dynamical splitting from the self-consistent induced magnetic field. Yet the zigzag triangle *should* be magnetic: its sublattice imbalance $N_A - N_B = 4$ (25 A -sites, 21 B -sites) produces **four zero-energy modes per spin**, and Lieb’s theorem predicts a ground state with total spin $S = |N_A - N_B|/2 = 2$.

2.2 The Hubbard interaction at mean-field level

Add the onsite Coulomb repulsion $H = H_{\text{TB}} + U \sum_i n_{i\uparrow} n_{i\downarrow}$. This term is quartic, so we treat it in **unrestricted Hartree–Fock (UHF)**:

$$n_{i\uparrow} n_{i\downarrow} \approx \langle n_{i\uparrow} \rangle n_{i\downarrow} + n_{i\uparrow} \langle n_{i\downarrow} \rangle - \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle, \quad (2)$$

which reduces the interaction to the spin-dependent onsite potential Eq. (1), $V_{i\sigma} = U(\langle n_{i,-\sigma} \rangle - \frac{1}{2})$ (particle-hole-symmetric form, keeping the spectrum symmetric about $E = 0$ at charge neutrality so $\mu = 0$). The up and down blocks are no longer identical: if a site has more down- than up-electrons, the up level is pushed up and the down level down, favouring further down-occupation — the **Stoner mechanism**. Above (here, any) critical U the symmetric solution is unstable

and the flake develops a site-resolved magnetisation $m_i = \frac{1}{2}(n_{i\uparrow} - n_{i\downarrow})$, $S_z = \sum_i m_i$, arranged antiferromagnetically (A-sublattice up, B down); the *net* moment is carried by the sublattice imbalance, giving $S_z = 2$ exactly (Lieb).

2.3 Self-consistency algorithm (solve_hubbard_mft)

$V_{i\sigma}$ depends on the occupations, which depend on $V_{i\sigma}$ — solved by iteration:

1. **Symmetry-breaking seed:** 2-colour the bond graph (BFS $\rightarrow s_i = \pm 1$) and start from a staggered guess $n_{i\uparrow} = \frac{1}{2} + \frac{1}{2}m_{\text{seed}}s_i$, $n_{i\downarrow} = \frac{1}{2} - \frac{1}{2}m_{\text{seed}}s_i$ (without a seed the symmetric solution is a fixed point and no magnetism appears).
2. Build H_{MF} by adding $U(n_{i,-\sigma} - \frac{1}{2})$ to the diagonal.
3. Diagonalise; **fill** (ρ_0), transform to the site basis, read off $n_{i\uparrow} = \rho_{ii}$, $n_{i\downarrow} = \rho_{N+i, N+i}$.
4. **Linear mix** $n \leftarrow (1 - \text{mix})n_{\text{old}} + \text{mix}n_{\text{new}}$ (damps the Stoner feedback so the loop converges).
5. Converge when $\max_i |\Delta n_{i\sigma}| < \text{tol}$ (10^{-8}), else iterate (up to 400).
6. Finalise: occupations, $V_{i\sigma}$, m_i , S_z , $\sum_i |m_i|$, and the HOMO–LUMO **spin gap**.

2.4 Where the field lives (and a subtlety fixed during development)

The converged $V_{i\sigma}$ is applied in **two** places so statics and dynamics see the same magnet:

- *Eigenproblem* / ρ_0 : folded into a copy $H_{\text{c,eig}}$, so the eigenvalues and the initial density matrix are magnetic.
- *Time evolution*: kept **out** of the tight-binding base H_0 and re-added on the diagonal *every step, in every branch* of `build_H_for_time`.

The second point matters: with `self_consistent_phase=true` the step rebuilds H from the hopping; if the exchange field lived only in H_0 that rebuild would drop it, ρ_0 would not be stationary, and it would churn at the $\sim \text{eV}$ exchange scale — wrong dynamics and a $\sim 4\times$ slowdown from tiny timesteps. Re-adding it every step keeps ρ_0 stationary.

2.5 The value of U

By default $U = v_{\text{ee}}(0) \approx 15.72 \text{ eV}$ — the onsite value of the graphene-specific fitted Coulomb kernel, the same number sitting on the V_{ee} diagonal. It is overridable from the config (`hubbard_U_eV`); the TOML key `coulomb_onsite_eV` is *not* used for U .

2.6 Configuration, outputs, validation

[features]		
hubbard	= true	# self-consistent UHF magnetism (needs spin_on = true)
# hubbard_U_eV	= 3.0	# optional U override in eV; unset/negative $\rightarrow v_{\text{ee}}(0) \sim 15.7 \text{ eV}$
hubbard_seed	= 0.5	# initial staggered moment for symmetry breaking
hubbard_mix	= 0.3	# linear mixing factor

Output: `magnetization.txt` (per-site x y sublattice n_{up} n_{dn} m_i plus a header with U , S_{total} , $\sum |m_i|$, gap, iterations) and a console summary. Validation on the reference flake: net $S_z = 2.000$ exactly (Lieb) at *every* U ; the moment magnitude and gap scale with U : Files: `Hamiltonians/hubbard.{hpp,cpp}` (new: `HubbardResult` + `solve_hubbard_mft`); `main.cpp` (run the solver, fold into $H_{\text{c,eig}}$, write

U (eV)	net S_z	$\sum_i m_i $	spin gap
15.72 ($v_{ee}(0)$)	2	19.44	13.41 eV
3.0	2	3.00	0.62 eV
0.5	2	2.11	0.085 eV

Table 1: The net $S = 2$ is Lieb-protected and survives to tiny U (the four exactly-degenerate zero modes spin-polarise for any repulsion — flat-band / Stoner ferromagnetism); the antiferromagnetic background $\sum |m_i|$ and the gap grow with U .

magnetization.txt); DensityMatrix/Density.{hpp,cpp} (re-add the field every step); params/* (the knobs). Fully documented in HUBBARD_FEATURE.md §1–9. With hubbard=false (default) none of this runs.

3 Implementation 2: time-dependent (TDUHF) spin channel

3.1 Goal and the key physical constraint

We want the local moments to respond to the pulse. A crucial fact shapes everything that follows: the Hamiltonian is **block-diagonal in spin** (spin-conserving hopping, the diagonal UHF potential, a σ_z Zeeman term, and a spin-blind dipole drive). Therefore the **total** $S_z = \frac{1}{2} \sum_i (n_{i\uparrow} - n_{i\downarrow})$ is exactly conserved. Light cannot change the net moment; it can only *redistribute* spin between sites, i.e. drive a transient **spin current** and local (de)magnetisation at fixed S_z .

3.2 Charge/spin decomposition (why there is no double counting)

Writing the onsite mean field in terms of the total charge $n_i = n_{i\uparrow} + n_{i\downarrow}$ and the moment $m_i = n_{i\uparrow} - n_{i\downarrow}$, Eq. (1) splits into a spin-blind *charge* part and a spin-antisymmetric *Stoner* part:

$$V_{i\sigma} = \underbrace{\frac{U}{2} \langle n_i \rangle}_{\text{charge}} \mp \underbrace{\frac{U}{2} \langle m_i \rangle}_{\text{spin (Stoner)}}, \quad (- \text{ for } \sigma = \uparrow, + \text{ for } \sigma = \downarrow). \quad (3)$$

The charge part is the same physical object as the onsite (v_{00}) piece of the classical Hartree term, which already couples to total charge and is spin-blind. **We therefore keep the Hartree term exactly as it was** — it is the charge-channel restoring force. Only the **spin part**, which couples to the moment and has *no* Hartree counterpart, is made dynamic. Because the two live in orthogonal channels, this cannot double-count.

3.3 What is added each step

On top of the frozen field we add the **live Stoner-exchange deviation**

$$\Delta V_{i,\uparrow/\downarrow}(t) = \mp \frac{U_{\text{spin}}}{2} (m_i(t) - m_i^{\text{eq}}), \quad m_i(t) = \rho_{ii}^{\uparrow}(t) - \rho_{ii}^{\downarrow}(t). \quad (4)$$

Two design points matter:

- It is a *functional of the instantaneous* $\rho(t)$, exactly like the Hartree term, so the adaptive RK (dopri5) stages carry the self-consistency: **no inner SCF loop**, $O(N)$ extra work per step.
- We drive it with the moment *deviation* $m_i(t) - m_i^{\text{eq}}$ on top of the frozen field, not the bare $m_i(t)$. This is the bare Stoner term plus a constant, and it guarantees two things: (a) **exact stationarity at any filling** — graphene edges are not locally half-filled, so anchoring on the frozen field keeps ρ_0 stationary for any site occupation; (b) $U_{\text{spin}} \rightarrow 0$ **reproduces the frozen dynamics exactly**. At exact half filling the two forms coincide.

U_{spin} (the Stoner coupling for the dynamic channel) is set separately from the charge U ; if unset it defaults to the charge U .

3.4 The instability that a naive version triggered, and the cure

A first attempt made the *full* UHF field dynamic and, to avoid double-counting the onsite Coulomb, deleted the onsite Hartree diagonal. Stripping that diagonal removed the charge-channel restoring force and the broken-symmetry state became **dynamically unstable** (a Thouless/RPA instability of time-dependent mean-field theory around a saddle). With *zero drive*, the dipole grew exponentially from numerical noise (e-folding ~ 6 a.u.), reaching $\mathcal{O}(10)$, and the adaptive stepper took $\sim 17\times$ more steps. The instability was worst at small U /small gap. The charge/spin split above cures it: the restoring force stays, and only the (bounded) spin channel is made dynamic.

3.5 Validation

On the reference flake at $U = 4.48$ eV (the previously divergent case):

zero-drive run	steps	max dipole	max spin induced
$U_{\text{spin}} = U$ (full dynamic spin)	1897	5×10^{-14}	3×10^{-8}
$U_{\text{spin}} = 0$ (reduces to frozen)	1948	1×10^{-8}	5×10^{-8}
<i>old scheme, same parameters</i>	~ 7700	<i>blows up to -9.9</i>	–

Table 2: Zero-drive stability. The ground state is now stationary to solver-noise level; the previously divergent case is stable and $\sim 4\times$ cheaper.

3.6 Configuration and files

[features]			
hubbard	= true	# required: self-consistent UHF magnetic ground state	
hubbard_dynamic	= true	# NEW: make the spin channel time-dependent (Stoner)	
# hubbard_U_spin_eV	= 4.48	# Stoner U for the dynamic spin channel;	
		# if unset/negative -> uses the charge U	

Files changed: `params/params.{hpp,cpp}` (new keys `hubbard_dynamic`, `hubbard_U_spin_eV`; derived `hub_U`, `hub_U_spin`, `hub_m_eq`); `main.cpp` (store the couplings and the equilibrium moment; console label); `DensityMatrix/Density.{hpp,cpp}` (solver carries the new fields; `build_H_for_time` adds the live Stoner deviation, the classical Hartree left untouched). With `hubbard_dynamic=false` the behaviour is byte-for-byte identical to before. Documented in `HUBBARD_FEATURE.md` §10.

4 Implementation 3: per-bond spin current

The code previously wrote only the *net* spin-current vector and the *charge* bond currents; there was no way to see spin flow in real space. We added a per-bond spin current, the spin-difference analogue of the existing charge bond current:

$$J_{\ell\ell'}^{\text{spin}} = \frac{2e}{\hbar} \text{Re}(H_{\ell\ell'}) [\text{Im} \rho_{\ell\ell'}^{\uparrow} - \text{Im} \rho_{\ell\ell'}^{\downarrow}], \quad (5)$$

using the same Peierls-phase convention as the charge current (relevant for the L2 self-consistent-phase runs). It is written every output step to `J_spin_bond_time_evolution.txt` (one column per bond) whenever `spin_on` and the system is 2D. Implementation: `compute_J_bonds_spin` in `Density.cpp`, streamed through the existing `RhoObserver` / `evolve_rho_over_time` plumbing.

5 What we measure, and the two drives

Because total S_z is conserved (§3), the informative spin observables are *not* the net moment (flat by construction) but:

- the **local induced moment** $m_i(t) - m_i^{\text{eq}}$ (where spin sits);
- the **staggered (Néel) moment** $m_{\text{stag}}(t) = \sum_i s_i m_i(t)$, with $s_i = \pm 1$ the sublattice label (the order parameter that can oscillate);
- the **spin current** J^{spin} (net magnitude and per-bond).

Two drive shapes are available and give the two response regimes:

- `mode = "ddf"`: a near-instantaneous dipole kick → **linear response** (susceptibility / extinction $\sigma(\omega)$);
- `mode = "time_impulse"`: a finite Gaussian-envelope carrier pulse → **nonlinear response** (high-harmonic generation, strong-field dynamics).

6 Figures and what they tell us

6.1 Control / validation: `plot_hubbard_dynamics.py`

Plots $S_z(t)$, $\sum_i |m_i|(t)$ and the per-site moments $m_i(t)$ from the induced spin diagonal. With *zero drive* all three are flat to $\sim 10^{-8}$: the magnetic ground state is a genuine stationary state of the TDUHF propagator. This is the sanity check behind the stability table above.

6.2 The U_{spin} dial: `plot_uspin_dial.py`

This figure is the direct demonstration that the dynamic spin channel is real: the exchange coupling builds a collective (Stoner/RPA) spin mode that is absent without it, while the charge channel is untouched — clean channel separation. Note the spectra are computed with the same Fourier convention as the built-in σ_{ext} but out to a higher cutoff, and weighted by ωIm so the DC component is suppressed and the resonances stand out (see §7 for why the built-in `sigma_ext.txt` alone, capped at `omega_cut_off`, missed these peaks).

6.3 The response matrix: `plot_response_matrix.py`

This is the “map” of the whole capability: the top row is the familiar charge optics; the bottom row is what the dynamic Hubbard makes accessible — a spin response in both the linear (a collective mode) and nonlinear (spin harmonics) regimes.

6.4 Light-driven spin current in real space: `plot_spin_current_realspace.py`

This is the physical picture the project is after: the pulse launches spin currents that shuttle spin density between sublattices (colour texture) while the net moment stays fixed — “redistribute, don’t pump”. The arrows and the colour are linked by continuity, $\partial_t m_i + \sum_{\text{bonds}} J^{\text{spin}} = 0$.

7 Physics findings

7.1 The gap and why the resonance moves so much with U

The non-interacting zigzag triangle has **four zero-energy modes per spin** (Lieb; $N_A - N_B = 4$), i.e. the single-particle gap at $U = 0$ is *zero*. Turning on U spin-polarises and splits these modes symmetrically about $E = 0$ by the exchange field; the eigenvalues cluster at $\pm 3.4\text{--}3.7\text{ eV}$ for

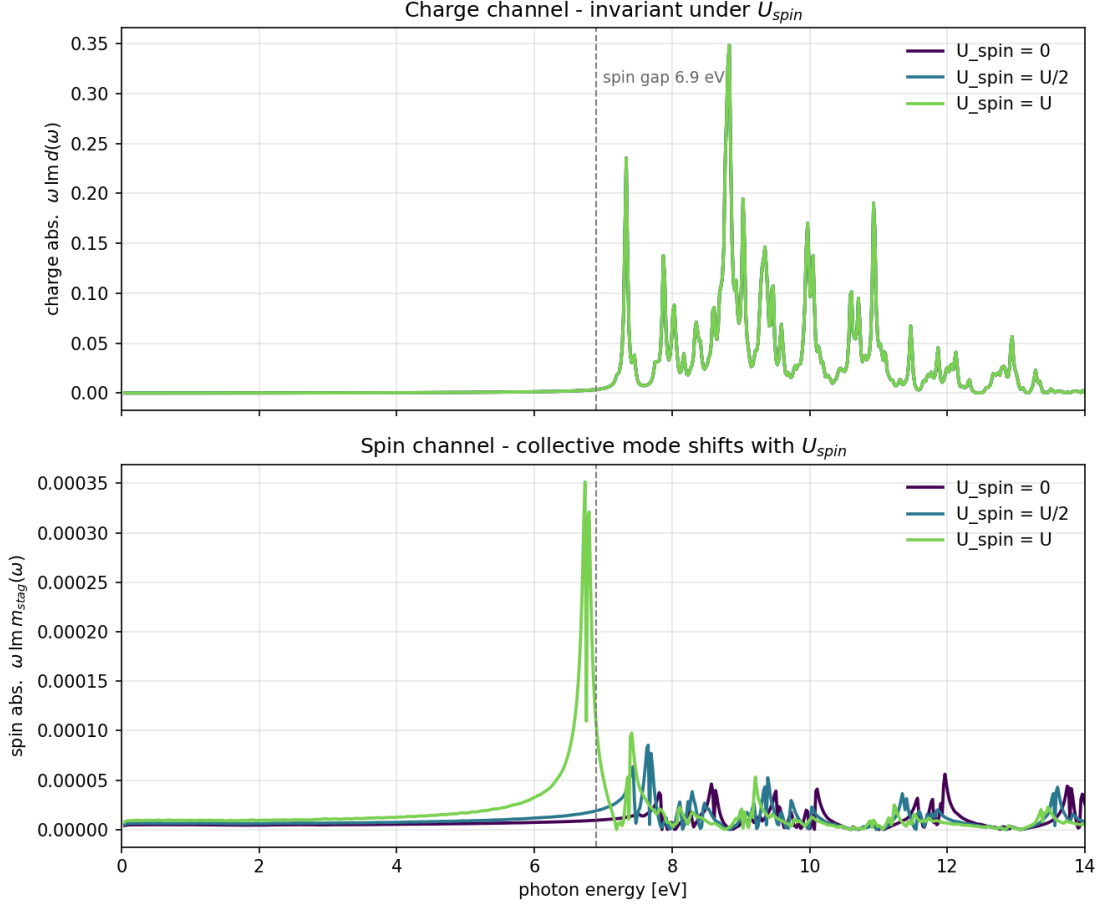


Figure 1: Linear-response (ddf) spectra for three Stoner couplings $U_{\text{spin}} = 0, U/2, U$ at fixed charge U . **Top:** charge extinction $\propto \omega \text{Im} d(\omega)$ — identical for all three (the charge channel does not see U_{spin}). **Bottom:** spin absorption $\propto \omega \text{Im} m_{\text{stag}}(\omega)$. A sharp collective spin mode appears at the spin gap (~ 6.9 eV) and *grows and sharpens with* U_{spin} ; at $U_{\text{spin}} = 0$ there is no gap peak, only the bare particle-hole background.

$U = 10.164$ eV, giving a HOMO–LUMO gap of 6.9 eV. The gap is therefore **entirely interaction-induced**, and tracks $\sim 2U\langle|m_i|\rangle$: Two cautions: (i) this is a mean-field **Slater** (symmetry-

U (eV)	$\sum_i m_i $	$\langle m \rangle$	$2U\langle m \rangle$	measured gap
4.48	4.03	0.088	0.79 eV	1.11 eV
10.164	15.05	0.327	6.65 eV	6.89 eV

Table 3: The gap is set by the self-consistent exchange splitting. Both U and the moment grow together, so the optical resonance shifts *faster* than linearly in U — a large shift is expected, and it appears for the frozen field too because it is a ground-state property.

broken) gap, *not* a true Mott gap — UHF systematically overestimates moments and gaps, so the magnitudes should not be read quantitatively; (ii) in the static ground state there is no double screening — the gap comes purely from the UHF field, the induced Hartree being zero at equilibrium.

7.2 Collective spin mode

The bottom panel of Fig. 1 shows the payoff of the dynamic channel: a collective spin resonance at the gap whose weight is controlled by U_{spin} . It is the time-dependent-mean-field (RPA) mode

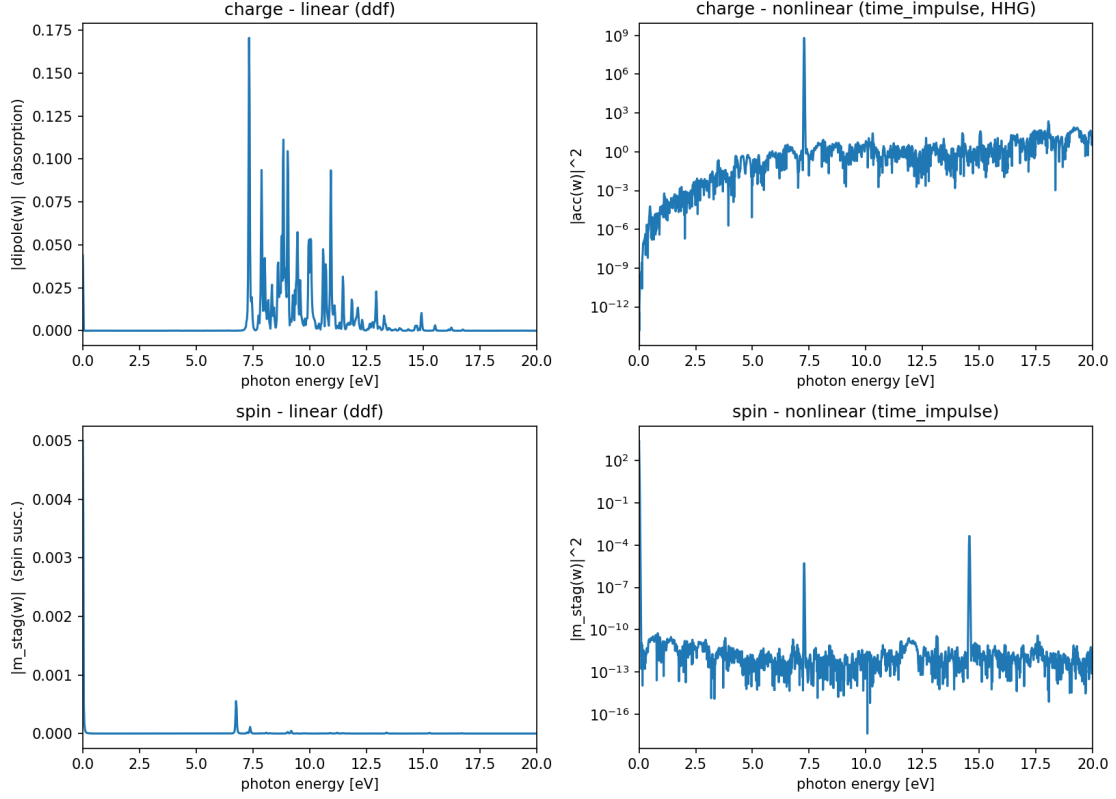


Figure 2: Two drives (columns: linear `ddf` / nonlinear `time_impulse`) $\times$ two channels (rows: charge / spin) in one view. Top row: charge absorption and charge HHG. Bottom row: the spin-channel analogues from m_{stag} .

of the broken-symmetry state, absent when the spin channel is frozen.

7.3 Spin transport at fixed S_z

Fig. 3 shows the central result enabled by both implementations together: an optical pulse drives real-space spin currents and a transient reduction of the Néel order, with the total spin exactly conserved.

8 How to reproduce

Reference flake: `configs/graphene_zigzag_triangle.toml` (5×5 zigzag triangle, `spin_on=true`, `hubbard=true`, `hubbard_dynamic=true`). Suggested run matrix:

run	drive	U_{spin}	purpose / figure
A	<code>ddf</code>	0	linear baseline (Fig. 1, Fig. 2)
B	<code>ddf</code>	$U/2, U$	U_{spin} dial (Fig. 1)
C	<code>time_impulse</code>	U	nonlinear + real-space spin current (Fig. 2, Fig. 3)

Practical notes:

- The `ddf` runs are cheap; only `time_impulse` runs are long.
- Keep U in the stable regime; very small gaps stress the mean-field dynamics.
- The adaptive solver writes non-uniform (and occasionally duplicated) time stamps — all spectral scripts resample onto a uniform grid before FFT.

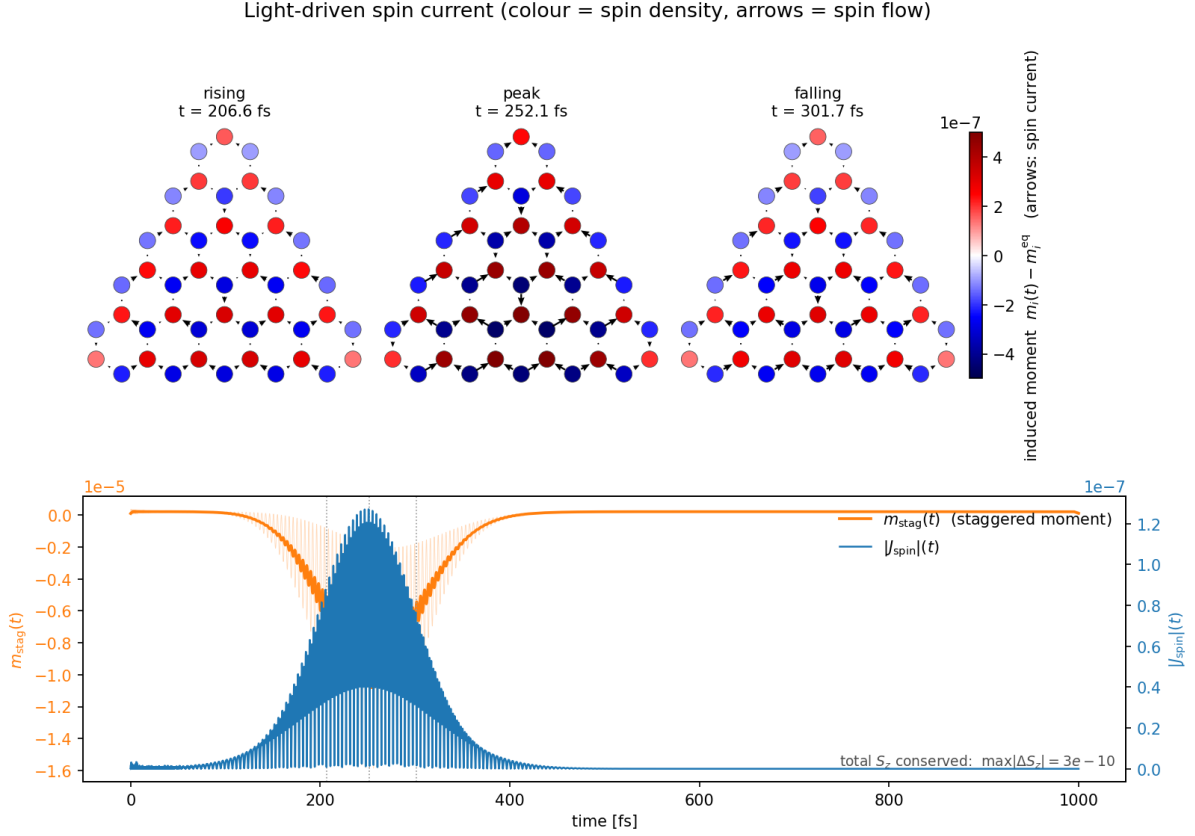


Figure 3: Strong-pulse (`time_impulse`) run. **Top:** three snapshots (rising / peak / falling of the response envelope). Node *colour* is the induced local moment $m_i(t) - m_i^{\text{eq}}$ (where spin has piled up/depleted); *black arrows* on the bonds are the per-bond spin current J^{spin} (where and how fast spin flows). **Bottom:** the staggered moment $m_{\text{stag}}(t)$ (a transient demagnetisation of the Néel order) and the net spin-current magnitude $|J_{\text{spin}}|(t)$ on twin axes; the annotation confirms total S_z is conserved to $\sim 10^{-8}$ – 10^{-10} .

- For the built-in `sigma_ext.txt` to contain the resonances, set `omega_cut_off` above the gap (the analysis scripts compute their own spectra to a higher cutoff regardless).
- `plot_spin_current_realspace.py` needs `J_spin_bond_time_evolution.txt`, i.e. a run made *after* the per-bond spin-current change.

Analysis scripts (all take a folder path at the top). `plot_hubbard_dynamics.py` (control), `plot_uspin_dial.py` (Fig. 1), `plot_response_matrix.py` (Fig. 2), `plot_spin_current_realspace.py` (Fig. 3).